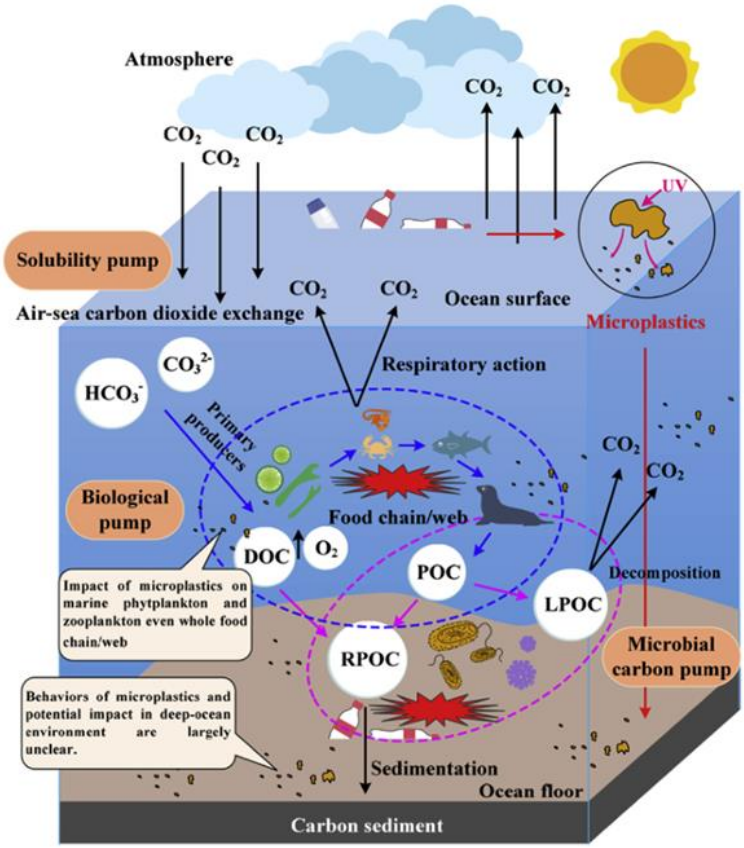
The Silent Collapse: The Impact of Warming Oceans on Fishermen

In downtown St. John's, fishing nets are pulled, but only 700,000 tons of wild salmon were caught, compared to 2.8 million tons of farmed salmon (Seafood Society, 2024). By 2048, seafood may vanish, leaving only jellyfish. It seems unbelievable, but this is a fact. Rising ocean temperatures is a major driver—over 90% of the past 50 years of warming has been absorbed by oceans (Lindsey & Dahlman, 2025). The livelihood of fishermen now depends on sustainable marine practices.

Unsustainable practices accelerate this crisis. First, overfishing reduces carbon sequestration, while trawling disturbs accumulated carbon stores. 0.34-0.37 Gt of carbon dioxide were released annually—equivalent to emissions from over 66 million cars (Atwood et al., 2024). Moreover, marine pollutants thickened the greenhouse veil. Rising sea temperatures also damage habitats, reducing fish populations. Bleaching-level heat stress scarred 84% of reefs globally and destroyed spawning grounds, leaving less polyps and shelter for zooxanthellae—the base of marine food. Fist population fell, endangering fishermen's survival worldwide. Without imminent actions, global temperatures will surpass the Paris Agreement threshold, pushing marine ecosystems towards collapse. (Hamilton and Feit, 2009, as cited in Chandrima Shaw et al., 2023).

Sustainable marine practices must be implemented both locally and globally. To begin with, the community, notably schools, could hold annual coastal clean-ups and recycle plastics to reduce greenhouse effects. Such collective actions are vital, since 79% of plastics end up in the environment (Geyer et al., 2017, cited in Shaw et al., 2023), releasing carbon dioxide and methane through photo-oxidation. More critically, the tiny microplastics pulverised from the estimated 640,000 tonnes of fishing gears lost in the oceans annually are made from synthetic plastics and accelerate the emission intensity. (Eurofish International Organisation, 2023). These initiatives foster environmental responsibility from a young age, with research stressing that children develop responsibility through activities that connect their actions to others (Dementiy & Grogoleva, 2016).



(Shen et al., 2020)

Building upon community efforts, governments could also enforce fishing bans to restore biological carbon pump. Increasing fish populations elevates carbon sequestrations, since fish store carbon and expel the rest as sinking pellets. Indeed, nearly 30% of all carbon sequestered by marine organisms is linked to fish (World Maritime University, 2025). Complementary measures incorporate establishing a local registration system for fishing vessels, limiting new entrants, maintaining sustainable fishing levels, and designating fisheries protection areas (Agriculture, Fisheries and Conservation Department, 2023). In these areas, restocking periodically could enable growth and reproduction to restore marine resources (Agriculture, Fisheries and Conservation Department, 2023). Furthermore, indigenous knowledges offer generational guidance for ecosystem restoration, since they include tracking, sustainable fishing and appropriate use of marine resources. Equally important, the government should prohibit trawling to reduce damage to the seabed and minimize carbon release. Instead, selective gears could be introduced - for instance, excluder leadings, which separate codends of varying mesh sizes (Feekings et al., 2023).

Additionally, artificial reef (AR) projects should continue to enhance fisheries resources. Monitoring surveys highlighted local ARs host higher abundance and diversity of fish than natural habitats, notably commercially valuable breams and snappers, which use them for shelter, feeding and reproduction (Agriculture, Fisheries and Conservation Department, 2023). Various types of AR may be deployed based on material availability, deployment purpose and site conditions (Agriculture, Fisheries and Conservation Department, 2023).

Most ARs, such as concrete blocks and old vessels, enhance habitat complexity. Others serve as biofilters to improve water quality or act as anti-trawling devices to protect nursery and spawning grounds. Taken together, these measures safeguard the livelihoods of coastal communities.

Admittedly, pro-development advocates may contend that sustainable marine practices reduce short-term profits for fishers. Nevertheless, by 2100, annual losses from climate change for just 16 major U.S. fisheries could range between $278 to 901 million (Moore et al., 2020). This demonstrates that without sustainable marine actions, we cannot guarantee human existence, fishermen's livelihoods, and economic stability in the long run. Therefore, fishermen should join the training sessions offered by local governments in transitioning toward sustainable marine practices, to adapt more effectively to new measures (Agriculture, Fisheries and Conservation Department, 2023). These sessions usually cover knowledges such as the use of selective fishing gears and recreational fishing. For policymakers, legislation against unsustainable practices, increased financial allocations into artificial reef projects and public encourage marine protections are essential. Ultimately, every individual should bear the responsibility to advocate for ocean conservations, including recycling plastics and supporting products labelled with marine sustainability.

The invisible, imminent impacts of climate change on vulnerable communities are often overlooked until we go off the boundary and are confronted with alarming numbers. The silent collapse is reversible, but only if we act now. Saving our oceans is not optional - it is a frontline of our fight against climate change and the safeguard of fishermen's livelihoods. If we all unite in marine sustainable actions, we can restore balance, protect resources, and ensure that the future generations inherit sustainable oceans rather than water filled only with jellyfish.

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